James Pham

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Education

University of British Columbia Vancouver, BC
Bachelor of Science in Computer Science *Expected Graduation: May 2028*

Awards
Trek Excellence Scholarship (Top 5% GPA in undergraduate year), Charles and Jane Banks Scholarship, Dean's Scholar

Experience

Software Developer Intern Jan. 2026 – Aug. 2026
Optum *Richmond, BC*

- Designed and implemented a comprehensive test suite for a new service, developing **unit, integration, and end-to-end tests** to validate functionality across multiple system components
- Developed **automated testing scripts in Python using Pytest** for a **GCP auditing service**, improving coverage and reliability of cloud-based auditing workflows while expanding automated test coverage
- Enhanced observability across **10+ JavaScript microservices** by implementing more granular **error logging**, improving debugging and issue diagnosis in distributed systems
- Containerized and automated the production deployment of a new service by developing **Docker and Jenkins configuration**, integrating the service into the team's **CI/CD pipeline**

Software Developer Sep. 2025 – Present
Motion UBC *Vancouver, BC*

- Developed a **full-stack** bottle pickup logistics platform, implementing user-facing features and integrating **Supabase backend services** to support core application functionality throughout the development lifecycle
- Implemented secure **login and account registration flows** using **Supabase Authentication**, managing **user sessions and authentication state** across the application

Software Developer Sep. 2026 – Aug. 2026
UBC Data Science Club *Vancouver, BC*

- Developed and maintained the club's primary website using **React**, designing **reusable, modular components** and implementing interactive user-facing features to improve functionality and maintainability
- Translated **Figma mockups** into **responsive interfaces**, implementing responsive layouts and reusable UI components while collaborating with the development team to deliver and iterate on new website features

Projects

CheetCode | cheetcode.com | *Python, WebSockets* February 2026 – Present

- Engineered a **real-time, room-based multiplayer coding platform** using **WebSockets**, implementing **synchronized game state** and **bidirectional communication** for concurrent users
- Designed and implemented a **Python backend** for **room creation, session management, and real-time event handling**, supporting concurrent multiplayer sessions and maintaining consistent state across connected clients
- Configured **CI/CD pipelines** for production deployment using **Vercel and Render**, managing build configurations, environment variables, and **frontend–backend integration** across development and production environments

Maze Generation and Solving Visualization Tool | *C++* April 2025 – Present

- Engineered an **interactive algorithm visualization tool** in **C++**, demonstrating maze generation and pathfinding through a graphical interface that animates execution for improved clarity and user understanding
- Implemented multiple **generation and search algorithms (e.g., DFS, BFS, A*)**, enabling real-time visual comparison while designing intuitive visualizations to highlight algorithm behavior and performance differences

Technical Skills

Languages: Python, C/C++, Java, SQL, JavaScript/TypeScript, HTML/CSS
Frameworks: Node.js, WebSockets, React, Next.js, PostgreSQL, Supabase
Developer Tools: Git, Docker, Google Cloud Platform, Jenkins, Postman, Jira